

VIKASH MADDI

☎ 8374501729 ✉ vikashmaddi@gmail.com 🔗 [linkedin.com/in/vikashmaddi](https://www.linkedin.com/in/vikashmaddi) 🐙 github.com/maddivikash

EDUCATION

Indian Institute of Technology Madras, Chennai, India

July 2018 – May 2022

Bachelor of Technology in Mechanical Engineering

8.31/10

PROFESSIONAL EXPERIENCE

VMock, Full Stack Developer

August 2022 – Present

Agentic Backend Services

- Built and operate a production **LLM** agent used by 5,000+ users, with a planning-first orchestration loop over a structured tool registry and Pydantic-validated schemas.
- Run the agent containerized on **Kubernetes**, maintaining long-lived stateful sessions and secure outbound egress for tool calls.

API Design & Microservices

- Cut API response latency from **400ms to 30ms** (13x) by adding targeted indexes, introducing **Redis** caching and refining service boundaries.
- Re-architected a monolithic utility module into a standalone **Python Flask** service with versioned **REST APIs**, providing a clear path for downstream teams.

Scalability & Reliability

- Led a zero-downtime cross-datacenter migration of 90k+ accounts across US and UK, using dual-write and hash-based reconciliation to achieve 99.9% data integrity.
- Operate containerized services on **Docker** and **Kubernetes** across **AWS** (S3, SQS) with end-to-end **CI/CD** pipelines from repository to production.
- Run event-driven workers over **AWS SQS** with explicit retry, timeout and idempotency semantics to guarantee reliable processing.

Observability & Tooling

- Stood up **LLM** observability with **Arize Phoenix**, instrumenting span-level traces so failures can be attributed to model, prompt, tool or harness.
- Built self-serve internal tooling and admin dashboards that automated user onboarding and tenant provisioning, removing 30% of manual operational work.

Karomi Technologies, Deep Learning Intern

May 2021 – August 2021

- Trained Faster R-CNN, RetinaNet and EfficientDet models in PyTorch with semi-supervised learning, reaching mAP of 96% for tables and 95% for symbols.

PROJECTS

Context Management Proxy and MCP Server | *Python, MCP, Agent Infrastructure*

2026

- Identified cost escalation in long-running agent tasks and built a transparent **HTTP proxy** and **MCP server** that compresses agent context without requiring client changes.
- Implemented relevance-based eviction and prefix-stable **prompt caching**; measured **cost** per completed task before and after, proving a measurable reduction.

TECHNICAL SKILLS

Languages: Python, TypeScript, SQL, JavaScript, Bash, C++, PHP, Java

Backend & APIs: FastAPI, Flask, Laravel, REST APIs, Microservices, asyncio, Pydantic, API Versioning

Containers & Cloud: Kubernetes, Docker, AWS (S3, SQS, IAM basics), Cloudflare Workers, Linux, Deployment Automation, Rollback Strategy

CI/CD & Tooling: GitHub Actions, Git-based Review, pytest, Self-Serve Internal Tooling, Policy-as-Code

Data & Retrieval: MySQL, PostgreSQL, Redis, Elasticsearch, Embeddings and Vector Retrieval, Chunking and Relevance Tuning, Event-Driven Pipelines, PyTorch

SCHOLASTIC ACHIEVEMENTS

- Secured a global rank of 38 in the July Long Challenge on the CodeChef platform.
- Secured an all-India open category rank of 1308 in JEE-ADVANCED 2018 among 1.8 million.